α-$BaCu_2Se_2$ Thin-Film XRD Analysis and Bayesian Optimization

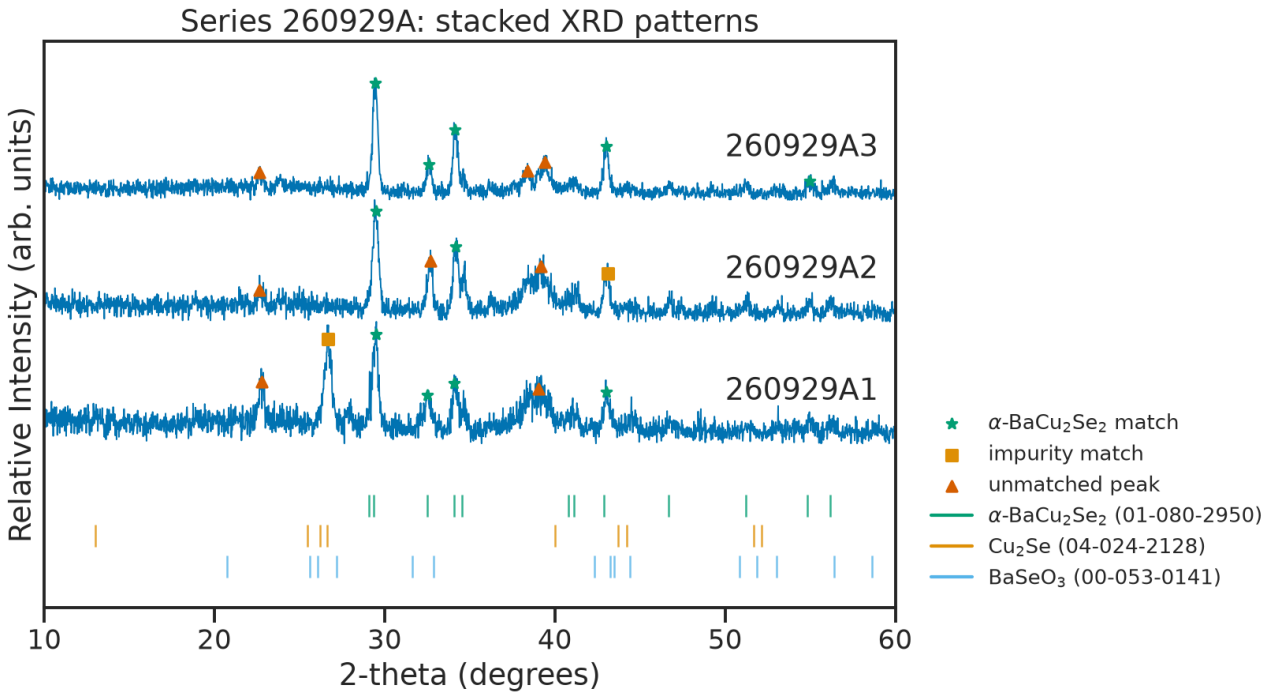
NOMAD-driven data extraction, phase matching, and GP-based synthesis planning

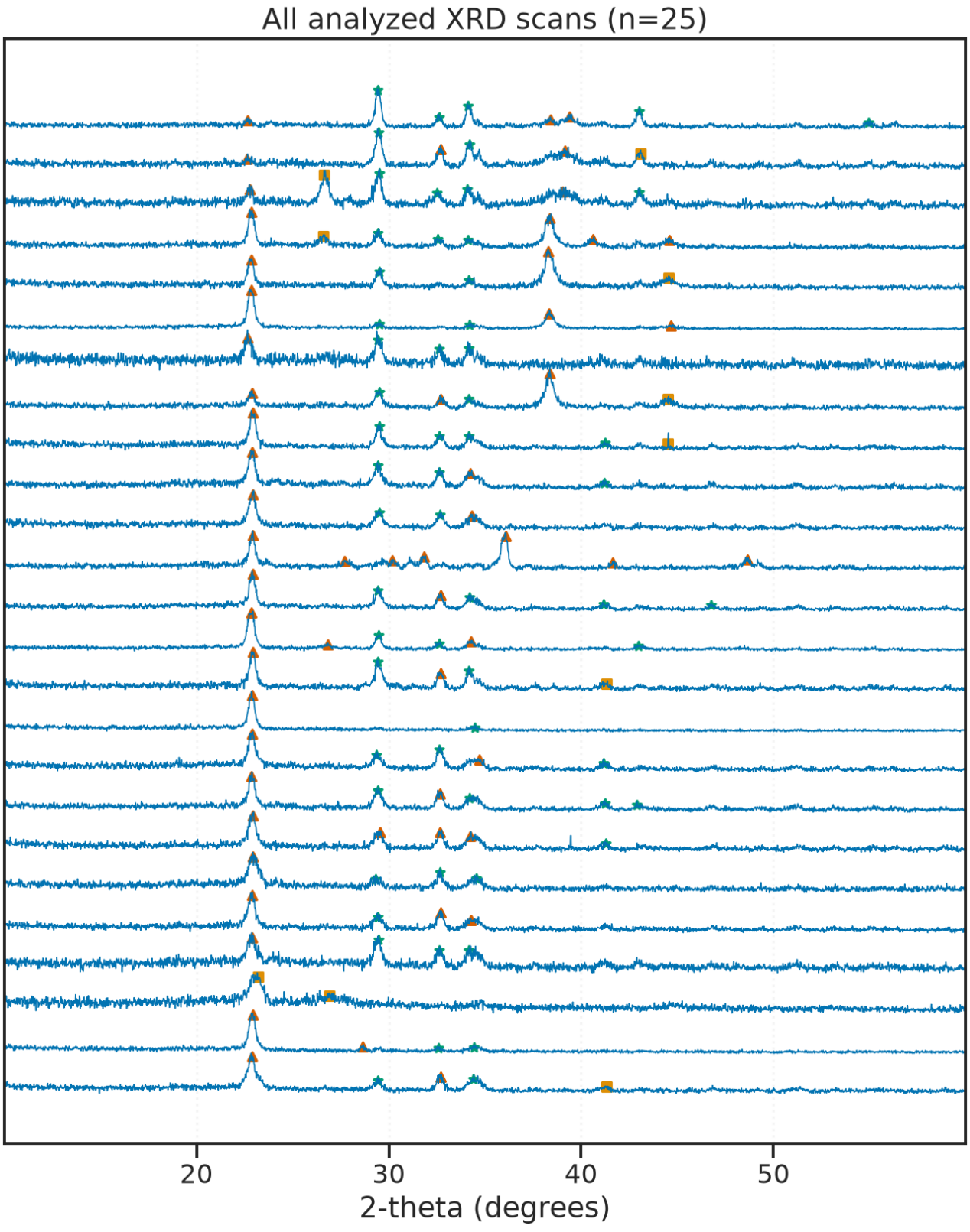
1. What was done

Synthesis records and XRD scans were pulled directly from the NOMAD Oasis REST API for the 42 α-$BaCu_2Se_2$ thin-film samples (INLThinFilmStack entries). Synthesis parameters (spin coating, soft/hard bake, selenization) and selenium-pulse settings were extracted for all samples; 35 include pulse on/off data. Each record also carries the precursor Cu/(Ba+Cu) ratio, the number of spin-coating cycles, the Mo-coating flag, and the spin-to-selenization day gap. For the 25 films with XRD scans, diffraction patterns were matched against ICDD reference cards for the α-$BaCu_2Se_2$ target (01-080-2950) and the Cu_2Se (04-024-2128) and $BaSeO_3$ (00-053-0141) impurities. Phase purity and crystallinity (1/FWHM of the strongest target match) were computed per film. A Gaussian-process Bayesian optimizer was trained on the 21 complete films to recommend the next synthesis conditions, optimizing objective = 0.7 × phase purity + 0.3 × normalized crystallinity over a 9-parameter space (spin speed, bake temperatures, selenization temperature/time, Se pulse on/off, Cu/(Ba+Cu) ratio, and the spin-to-selenization day gap).

2. XRD phase identification

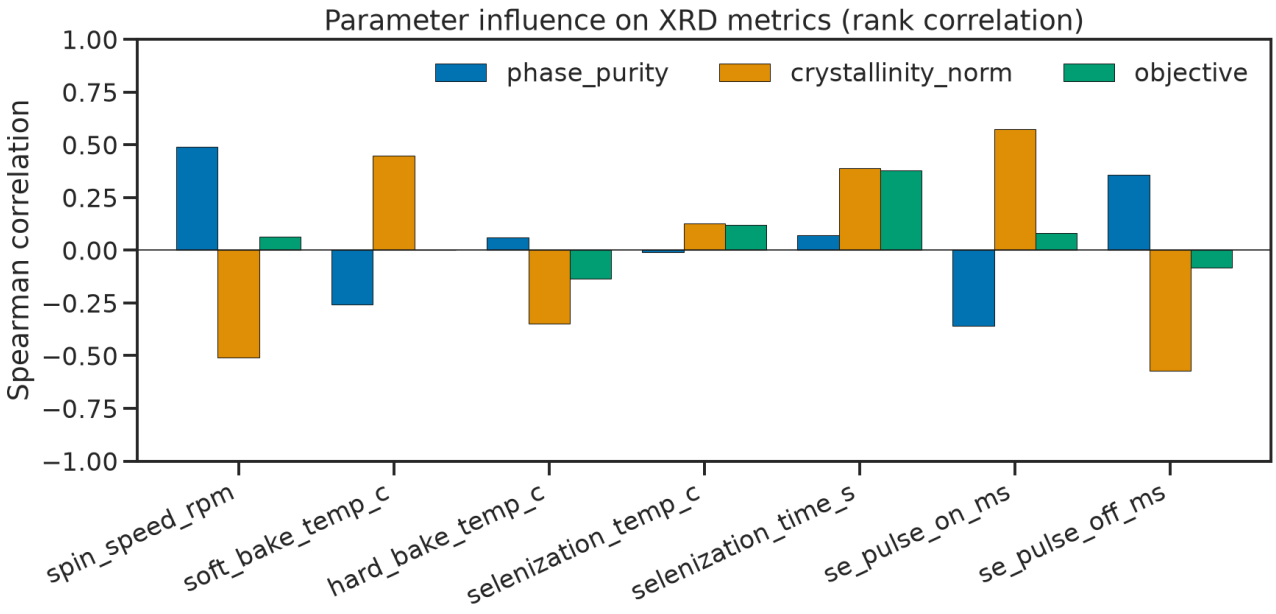
Stacked XRD patterns per series with reference bars for the target and impurity phases. Markers flag target (star), impurity (square), and unmatched (triangle) peaks.

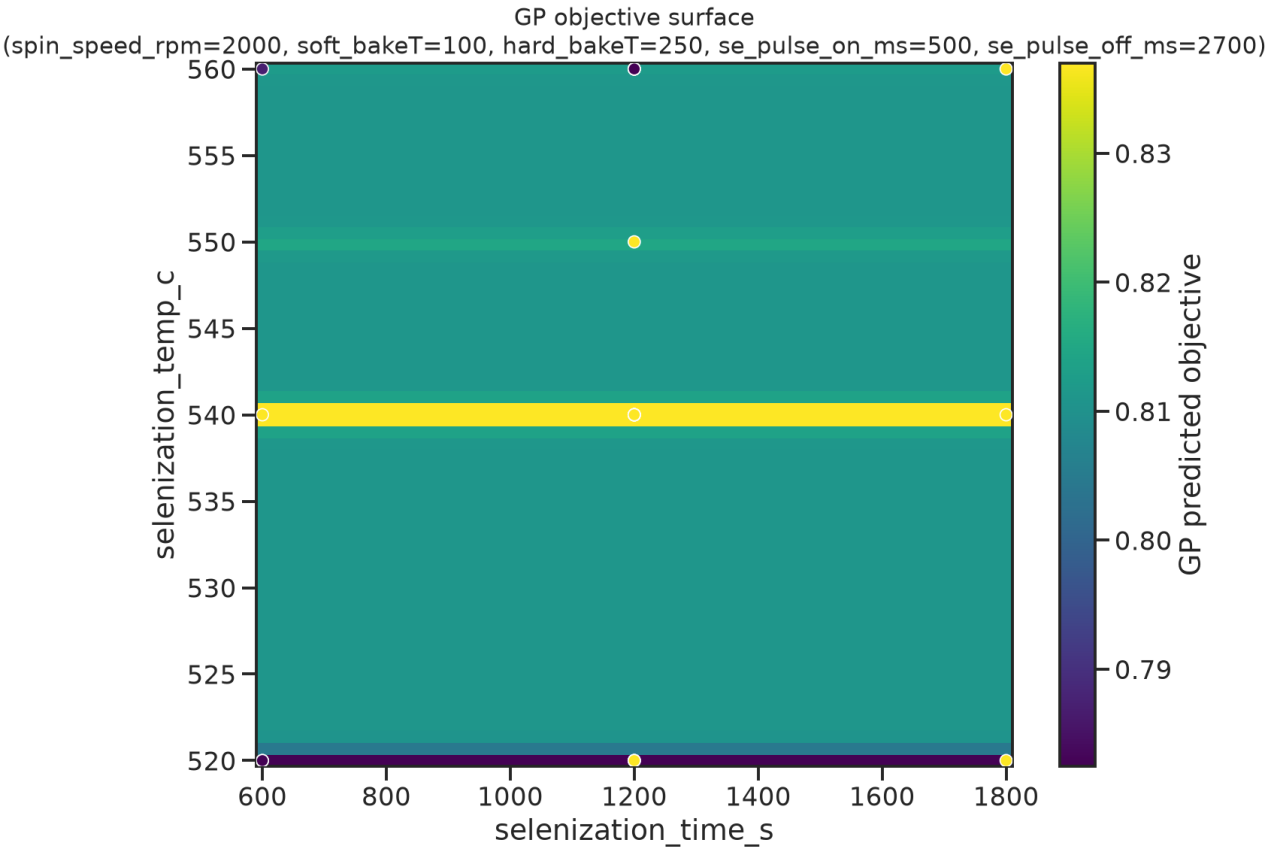




3. Main results

Across the analyzed films the mean phase purity is 0.91 and mean crystallinity is 2.18. The best film is 260429B2 (objective 1.00; purity 1.00, crystallinity 2.92). Rank (Spearman) correlation with the objective identifies selenization time as the dominant synthesis lever; bake temperatures and pulse timing are weaker.



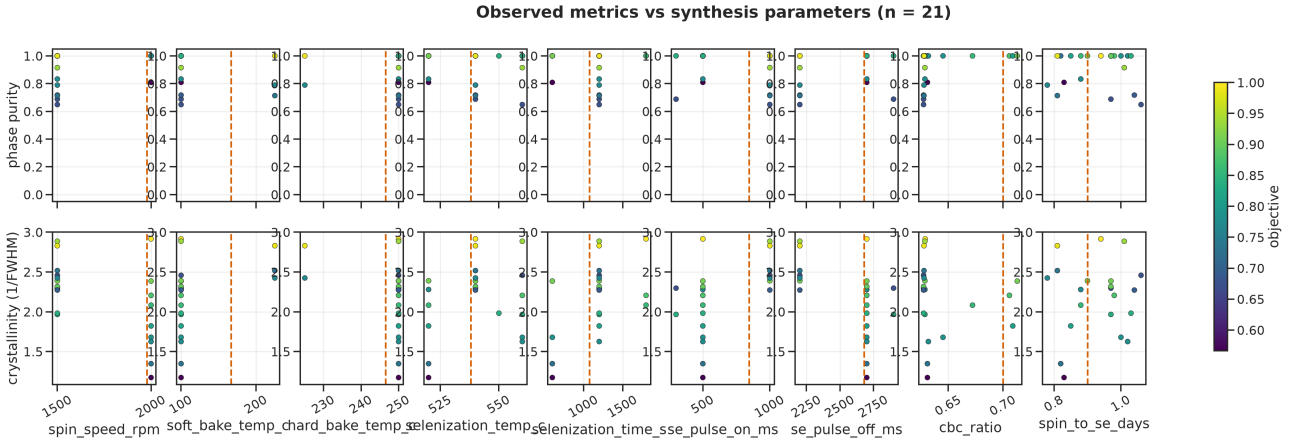


4. Suggested next experiments

Top Gaussian-process suggestions (predicted objective ± std), ranked by expected improvement:

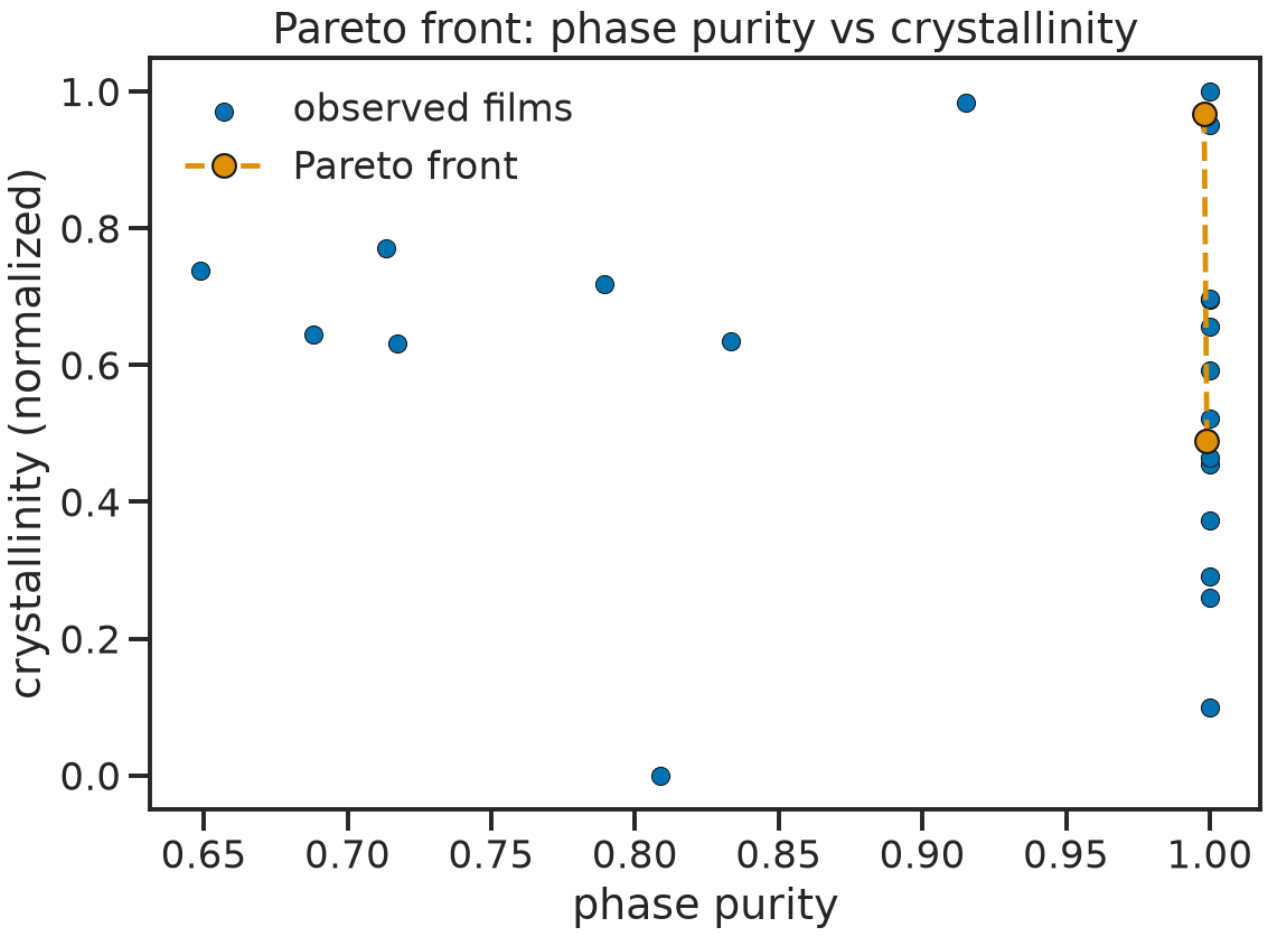
rank	spin (s)peed rpm	soft bake temp (°C)	hard bake temp (°C)	selenization temp (°C)	selenization time (s)	se pulse on (ms)	se pu
1	1978	166	246	538	1073	846	
2	1832	170	246	538	994	926	
3	1853	180	246	547	877	673	
4	2000	116	250	543	919	666	
5	2000	218	249	552	1222	405	

Recommended first experiment: spin 1978 rpm, soft bake 166 °C, hard bake 246 °C, selenization 538 °C for 1073 s with Se pulse 846 ms on / 2680 ms off (predicted objective 0.81).



5. Multi-objective optimization (Ax)

A complementary multi-objective search treats phase purity and normalized crystallinity as two separate objectives instead of one weighted score. Ax (BoTorch) fits a Gaussian process per metric and finds the Pareto front via qNEHVI hypervolume maximization. The suggested conditions below are Pareto-optimal candidates spanning the purity–crystallinity trade-off.



rank	spin (s)peed rpm	soft bake temp (°C)	hard bake temp (°C)	selenization temp (°C)	selenization time (s)	se pulse on (ms)	se pu
1	1916	183	247	539	1791	601	
2	2000	100	250	544	1800	659	

rank	spin (s)peed rpm	soft bake temp (°C)	hard bake temp (°C)	selenization temp (°C)	selenization time (s)	se pulse on (ms)	se pu
3	2000	100	250	537	1800	399	
4	1842	100	238	538	1800	596	
5	2000	197	238	545	1800	434	